

Graphical method.

1) Z-equation line method

2) Best of all corners method

1) Z-equation line method

→ Write the objective function

$$Z = C_1x + C_2y$$

→ Choose a convenient value for Z and draw a line

→ This line can be shifted parallelly outward (for maximization) or inward (for minimization)

→ The last point where the line touches the feasible region is optimum solution

This method is more visual but less convenient for exact numerical answers.

1) Maximize $Z = 3x_1 + 5x_2$

Subjected to

$$x_1 + 2x_2 \leq 2000$$

$$x_1 + x_2 \leq 1500$$

$$x_2 \leq 600$$

$$x_1, x_2 \geq 0$$

Solⁿ:-

$$x_1 + 2x_2 = 2000 \quad \text{--- ①}$$

$$x_1 + x_2 = 1500 \quad \text{--- ②}$$

$$x_2 = 600 \quad \text{--- ③}$$

Put $x_1 = 0$ in ① $\Rightarrow x_2 = 1000$

Put $x_2 = 0$ in ① $\Rightarrow x_1 = 2000$

$$\therefore (x_1, x_2) = (2000, 1000)$$

Put $x_1 = 0$ in ② $\Rightarrow x_2 = 1500$

Put $x_2 = 0$ in ② $\Rightarrow x_1 = 1500$

$$\therefore (x_1, x_2) = (1500, 1500)$$

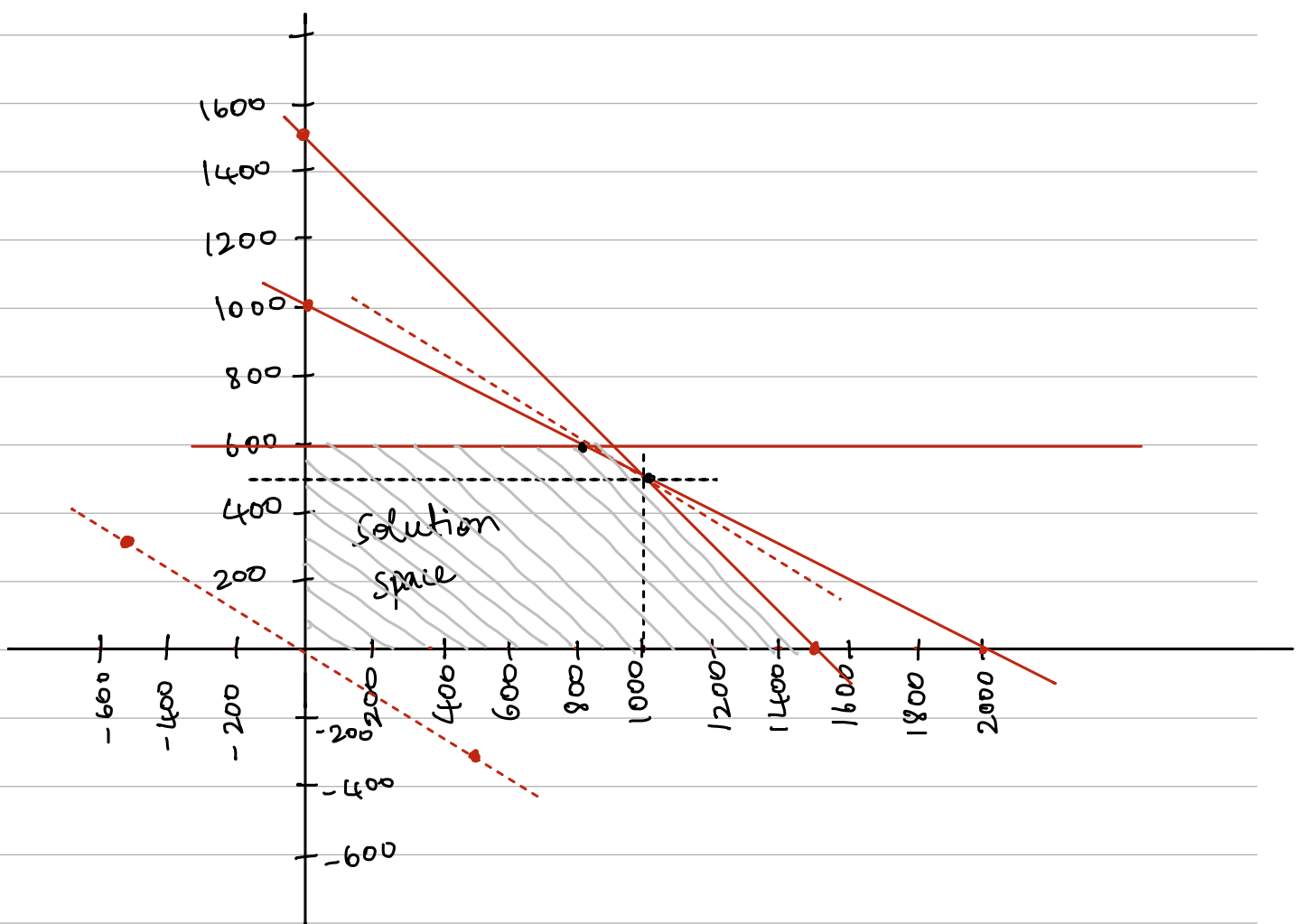
Eqn ③ $\Rightarrow x_2 = 600, x_1 = 0$

$$\therefore (x_1, x_2) = (0, 600)$$

Put $Z = 3x_1 + 5x_2 = 0$

$$\Rightarrow \frac{x_1}{x_2} = -\frac{5}{3}$$

points are $(-500, 300)$, $(500, -300)$



$$Z = 3x_1 + 5x_2 = 3(1000) + 5(500) = 5500$$

The values are $x_1 = 1000$, $x_2 = 500$ and $Z = 5500$.

2. Minimize $Z = 20x_1 + 10x_2$, Subjected to

$$x_1 + 2x_2 \leq 40$$

$$3x_1 + x_2 \geq 30$$

$$4x_1 + 3x_2 \geq 60$$

$$x_1, x_2 \geq 0$$

Solⁿ:- $x_1 + 2x_2 = 40$ — (1)

$$3x_1 + x_2 = 30$$
 — (2)

$$4x_1 + 3x_2 = 60$$
 — (3)

$$x_1 = 0 \text{ in (1)} \Rightarrow x_2 = 20 \quad (40, 20)$$

$$x_2 = 0 \text{ in (1)} \Rightarrow x_1 = 40$$

$$x_1 = 0 \text{ in (2)} \Rightarrow x_2 = 30 \quad (10, 30)$$

$$x_2 = 0 \text{ in (2)} \Rightarrow x_1 = 10$$

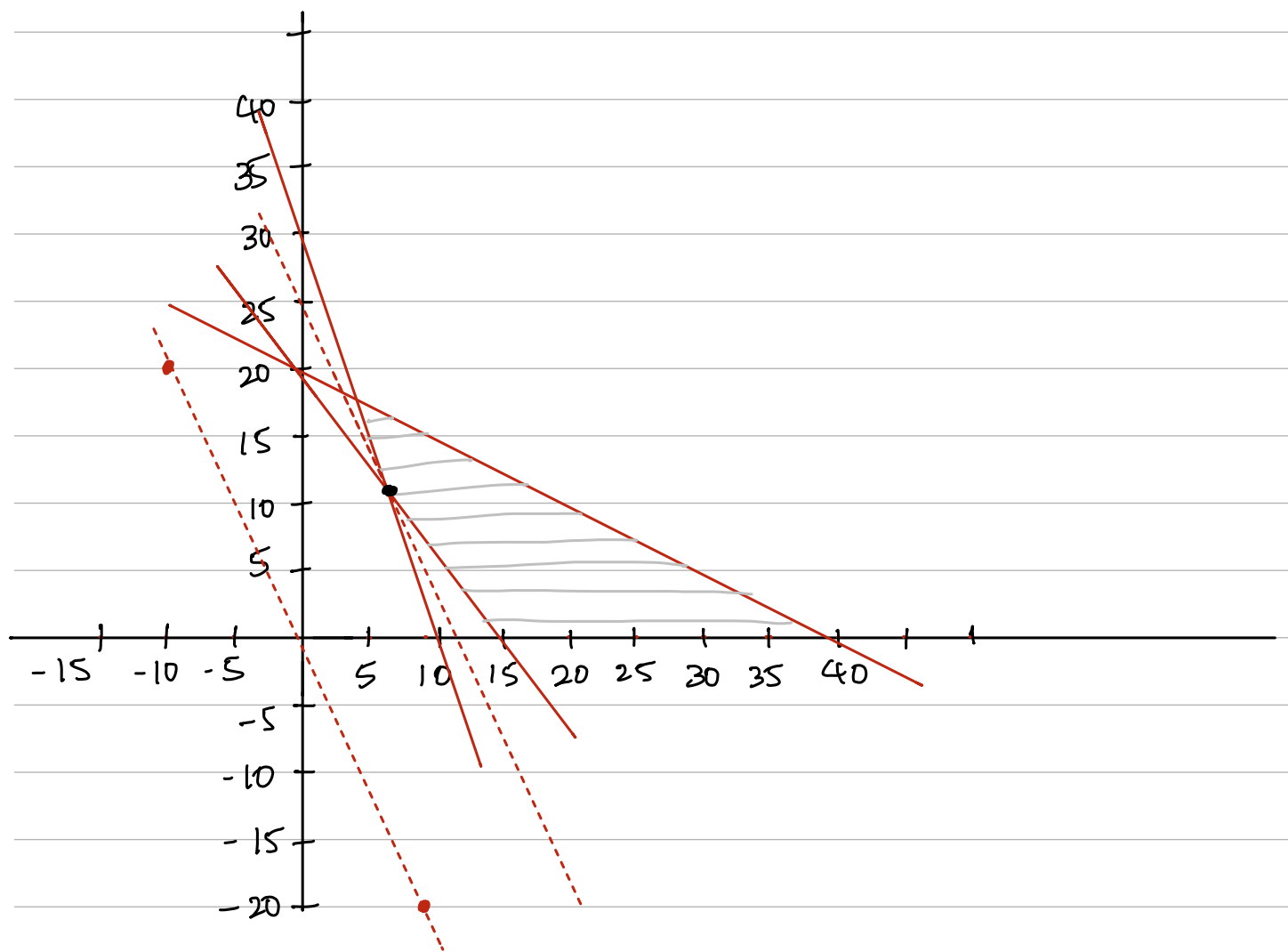
$$x_1 = 0 \text{ in (3)} \Rightarrow x_2 = 20 \quad (15, 20)$$

$$x_2 = 0 \text{ in (3)} \Rightarrow x_1 = 15$$

Consider, $20x_1 + 10x_2 = 0$

$$\Rightarrow \frac{x_1}{x_2} = \frac{-10}{20}$$

points are $(-10, 20)$ & $(10, -20)$



$\therefore$ point of intersection is intersection of
eqn ② & ③

$$3x_1 + x_2 = 30$$

$$4x_1 + 3x_2 = 60$$

$$\Rightarrow x_1 = 6 \text{ and } x_2 = 12$$

$$\Rightarrow Z = 20x_1 + 10x_2 = 20(6) + 10(12) = 240$$

Redundant constraint

A constraint that does not affect the feasible region of the problem.

Multiple optimal solution

More than one optimal solution

Infeasibility

No solution which satisfies all the constraints simultaneously.

Simplex method.

1. Maximize $Z = 3x_1 + 5x_2 + 4x_3$

Subject to

$$2x_1 + 3x_2 \leq 8$$

$$2x_2 + 5x_3 \leq 10$$

$$3x_1 + 2x_2 + 4x_3 \leq 15,$$

$$x_1, x_2 \geq 0$$

Soln:-

$$2x_1 + 3x_2 + s_1 = 8$$

$$2x_2 + 5x_3 + s_2 = 10$$

$$3x_1 + 2x_2 + 4x_3 + s_3 = 15$$

$$x_1 \geq 0, x_2 \geq 0, s_1, s_2, s_3 \geq 0$$

$$\Rightarrow Z = 3x_1 + 5x_2 + 4x_3 + 0s_1 + 0s_2 + 0s_3$$

	C_j	3	5	4	0	0	0		
C_B	Basic variable	x_1	x_2	x_3	s_1	s_2	s_3	RHS	Ratio
0	s_1	2	3	0	1	0	0	8	$8/3 \rightarrow \min$
0	s_2	0	2	5	0	1	0	10	$10/2$
0	s_3	3	2	4	0	0	1	15	$15/2$
Z_j		0	0	0	0	0	0	0	
$C_j - Z_j$		3	5	4	0	0	0		

↓
max value

$$Z_j = \sum_{i=1}^3 C B_j (a_{ij})$$

$$Z_1 = \sum_{i=1}^3 C B_j (a_{ij})$$

$$= C B_1(a_{11}) + C B_2(a_{12}) + C B_3(a_{13})$$

$$= (0)(2) + (0)(3) + (0)(0) = 0$$

$$Z_2 = 0$$

$$Z_3 = 0$$

Key row $\rightarrow S_1$, Key column $\rightarrow x_2$

Key element is 3.

Iteration I

$C B_i$	C_j	3	5	4	0	0	0		
	B.V	x_1	x_2	x_3	S_1	S_2	S_3	RHS	Ratio
5	x_2	$2/3$	$2/3$	$0/3$	$1/3$	$0/3$	$0/3$	$8/3$	—
0	S_2	$-4/3$	0	5	$-2/3$	1	0	$14/3$	$14/15 \rightarrow \min$
	S_3	$5/3$	0	4	$-2/3$	0	1	$29/3$	$29/12$
	Z_j	$10/3$	$15/3$	0	$5/3$	0	0	$40/3$	—
	$C_j - Z_j$	$-1/3$	0	4	$-5/3$	0	0		

↓
max

$$\text{New value} = \text{Old value} - \frac{\text{Corr. key column value} \times \text{Corr. key row value}}{\text{Key element}}$$

New values

$$0 - \frac{2 \times 2}{3} = -\frac{4}{3}$$

$$3 - \frac{2 \times 2}{3} = \frac{5}{3}$$

$$2 - \frac{2 \times 3}{3} = 0$$

$$2 - \frac{2 \times 3}{3} = 0$$

$$5 - \frac{2 \times 0}{3} = 5$$

$$4 - \frac{2 \times 0}{3} = 4$$

$$0 - \frac{2 \times 1}{3} = -\frac{2}{3}$$

$$0 - \frac{2 \times 1}{3} = -\frac{2}{3}$$

$$1 - \frac{2 \times 0}{3} = 1$$

$$0 - \frac{2 \times 0}{3} = 0$$

$$0 - \frac{2 \times 0}{3} = 0$$

$$1 - \frac{2 \times 0}{3} = 1$$

$$10 - \frac{2 \times 8}{3} = \frac{14}{3}$$

$$15 - \frac{8 \times 2}{3} = \frac{29}{3}$$

Iteration II

	C_j	3	5	4	0	0	0		
$C B_i$	BV	x_1	x_2	x_3	s_1	s_2	s_3	RHS	Ratio
5	x_2	$\frac{2}{3}$	1	0	$\frac{1}{3}$	0	0	$\frac{8}{3}$	4
4	x_3	$-\frac{4}{15}$	0	$\frac{5}{5}$	$-\frac{2}{15}$	$\frac{1}{5}$	0	$\frac{14}{15}$	-4 skip $\frac{7}{14}$
0	s_3	$\frac{41}{15}$	0	0	$-\frac{2}{15}$	$-\frac{4}{3}$	1	$\frac{89}{15}$	$\frac{89}{41} \rightarrow \min$
	Z_j	$\frac{34}{15}$	5	4	$\frac{17}{15}$	$\frac{4}{5}$	0	$\frac{256}{15}$	
	$C_j - Z_j$	$\frac{11}{5}$	0	0	$-\frac{17}{15}$	$-\frac{4}{5}$	0		

New value

$$\frac{5}{3} + \frac{16}{3 \times 5} = \frac{41}{15}$$

$$0 - \frac{0}{5} = 0$$

$$4 - \frac{4 \times 5}{5} = 0$$

$$-\frac{2}{3} + \frac{8}{15} = -\frac{2}{15}$$

$$0 - \frac{4 \times 1}{5} = -\frac{4}{5}$$

$$1 - \frac{0}{5} = 1$$

$$\frac{29}{3} - \frac{4 \times 14}{3 \times 5} = \frac{89}{15}$$

$$\frac{1}{3} + \frac{2/3 \times 2/15}{4/15}$$

$$\frac{1}{3} + \frac{4}{3 \times 41}$$

Iteration III

	C_j	3	5	4	0	0	0		
CB _j	BV	x_1	x_2	x_3	s_1	s_2	s_3	RHS	Ratio
5	x_2	0	1	0	$15/41$	$8/41$	$-10/41$	$50/41$	
4	x_3	0	0	1	$-6/41$	$5/41$	$4/41$	$62/41$	
3	x_1	1	0	0	$-2/41$	$-12/41$	$15/41$	$89/41$	
	Z_j	3	5	4	$45/41$	$24/41$	$11/41$	$765/41$	
	$C_j - Z_j$	0	0	0	$-45/41$	$-24/41$	$-11/41$		

$$1) \frac{2}{3} - \frac{2}{3} \times \frac{41}{15}$$

$$\frac{5 \times 15}{41} +$$

$$2) 1 - 2/3 \times 0$$

$$\frac{15}{41} + \frac{8}{41} - \frac{10}{41}$$

$$3) 0 - 2/3 \times 0$$

$$33 - 1 \frac{23}{41}$$

∴ Solution is

Basic variables

$$x_1 = 89/41 \quad x_2 = 62/41 \quad x_3 = 50/41$$

Objective value

$$Z = \frac{765}{41}$$

	C_j	2	3	0	0		
C_B	B.V	x_1	x_2	s_1	s_2	RHS	Ratio
0	s_1	2	1	1	0	12	12
0	s_2	1	(3)	0	1	15	$15/3 \rightarrow \min$
	Z_j	0	0	0	0	0	—
	$C_j - Z_j$	2	3	0	0		

↓
max

	C_j	2	3	0	0		
	B.V	x_1	x_2	s_1	s_2	RHS	Ratio
0	s_1	($5/3$)	0	1	$-1/3$	7	$21/5 \rightarrow \min$
3	x_2	$1/3$	1	0	$1/3$	$15/3$	15
	Z_j	1	3	0	1	15	
	$C_j - Z_j$	1	0	0	-1		

↑
max

	C_j	2	3	0	0	
	B.V	x_1	x_2	s_1	s_2	RHS
2	x_1	1	0	$3/5$	$-1/5$	$21/5$
3	x_2	0	1	$-1/5$	$6/15$	$18/5$
	Z_j	2	3	$3/5$	$12/15$	$96/5$
	$C_j - Z_j$	0	0	$-3/5$	$-12/15$	

$$x_1 = 21/5 \quad x_2 = 18/5$$

$$Z_{\max} = 96/5$$

Minimization problem

Minimize $Z = x_1 - 3x_2 + 3x_3$ Subject to

$$3x_1 - x_2 + 2x_3 \leq 7$$

$$2x_1 + 4x_2 \geq -12$$

$$-4x_1 + 3x_2 + 8x_3 \leq 10$$

Solⁿ:- LPP in standard form.

Maximize $Z^* = -x_1 + 3x_2 - 3x_3$
Subject to

$$3x_1 - x_2 + 2x_3 + s_1 = 7$$

$$-2x_1 - 4x_2 + s_2 = 12$$

$$-4x_1 + 3x_2 + 8x_3 + s_3 = 10$$

$$x_1, x_2, x_3, s_1, s_2, s_3 \geq 0$$

C_j		-1	3	-3	0	0	0		
B.V		x_1	x_2	x_3	s_1	s_2	s_3	RHS	Ratio
0	s_1	3	-1	2	1	0	0	7	-
0	s_2	-2	-4	0	0	1	0	12	-
0	s_3	-4	3	8	0	0	1	10	10/3
Z_j		0	0	0	0	0	0	0	
$C_j - Z_j$		-1	3	-3	0	0	0		

C_j		-1	3	-3	0	0	0		
B.V		x_1	x_2	x_3	s_1	s_2	s_3	RHS	Ratio
0	s_1	$\frac{5}{3}$	0	$\frac{14}{3}$	1	0	$\frac{1}{3}$	$3\frac{1}{3}$	$3\frac{1}{5}$
0	s_2	$-\frac{22}{3}$	0	$3\frac{2}{3}$	0	1	$\frac{4}{3}$	$7\frac{2}{3}$	—
3	x_2	$-\frac{4}{3}$	1	$\frac{8}{3}$	0	0	$\frac{1}{3}$	$10\frac{1}{3}$	—
Z_j		-4	3	8	0	0	1	10	
$C_j - Z_j$		3	0	-11	0	0	-1		

C_j		-1	3	-3	0	0	0	Sol^n
B.V		x_1	x_2	x_3	s_1	s_2	s_3	
-1	x_1	1	0	$\frac{14}{5}$	$\frac{3}{5}$	0	$\frac{1}{5}$	$3\frac{1}{5}$
0	s_2	0	0	$15\frac{6}{5}$	$2\frac{2}{5}$	1	$\frac{14}{5}$	$35\frac{4}{5}$
3	x_2	0	1	$3\frac{2}{5}$	$\frac{4}{5}$	0	$\frac{3}{5}$	$58\frac{1}{5}$
Z_j		-1	3	$8\frac{2}{5}$	$\frac{9}{5}$	0	$\frac{8}{5}$	$14\frac{3}{5}$
$C_j - Z_j$		0	0	$-9\frac{7}{5}$	$-\frac{9}{5}$	0	$-\frac{8}{5}$	

As $C_j - Z_j \leq 0$, The solution is optimal.

$\therefore$ The optimal solⁿ is $Z^* = \frac{143}{5}$

$$\Rightarrow Z_{\min} = -\frac{143}{5}$$

$$x_1 = 3\frac{1}{5} \quad x_2 = 58\frac{1}{5} \quad x_3 = 0$$

Minimize $Z = x_1 - 3x_2 + 3x_3$ Subject to

$$3x_1 - x_2 + 2x_3 \leq 7$$

$$2x_1 + 4x_2 \geq -12$$

$$-4x_1 + 3x_2 + 8x_3 \leq 10$$

Standard form

Minimize $Z = x_1 - 3x_2 + 3x_3$

s.t $3x_1 - x_2 + 2x_3 + S_1 = 7$

$$-2x_1 - 4x_2 + S_2 = 12$$

$$-4x_1 + 3x_2 + 8x_3 + S_3 = 10$$

	C_j	1	-3	3	0	0	0		
CB _i	B.V	x_1	x_2	x_3	S_1	S_2	S_3	RHS	Ratio
0	S_1	3	-1	2	1	0	0	7	-
0	S_2	-2	-4	0	0	1	0	12	-
0	S_3	-4	3	8	0	0	1	10	10/3
	Z_j	0	0	0	0	0	0	0	
	$C_j - Z_j$	1	-1	3	0	0	0		

↓
max negative

$$R_3 \rightarrow R_3/3 \quad R_1 \rightarrow R_1 - \frac{(-1)(R_3)}{3} = R_1 + \frac{R_3}{3}$$

$$R_2 \rightarrow R_2 - \frac{(-4)(R_3)}{3} = R_2 + \frac{4R_3}{3}$$

	C_j	1	-3	3	0	0	0		
CB _i	B.V	x_1	x_2	x_3	s_1	s_2	s_3	RHS	Ratio
0	s_1	$5/3$	0	$14/3$	1	0	$1/3$	$31/3$	$31/5$
0	s_2	$-22/3$	0	$32/3$	0	1	$4/3$	$76/3$	—
-3	x_2	$-4/3$	1	$8/3$	0	0	$1/3$	$10/3$	—
	Z_j	4	-3	-8	0	0	-1	-10	
	$C_j - Z_j$	-3	0	11	0	0	1		
		↓							
		max negative							

$$R_1 \rightarrow R_1 / 5/3 = \frac{3R_1}{5}$$

$$R_2 \rightarrow R_2 - \frac{\left(-\frac{22}{3}\right)(R_1)}{\left(5/3\right)} = R_2 + \frac{22}{5} R_1$$

$$R_3 \rightarrow R_2 - \frac{\left(-4/3\right)(R_1)}{\left(5/3\right)} = R_2 + 4/5 R_1$$

	C_j	1	-3	3	0	0	0		
CB _i	B.V	x_1	x_2	x_3	s_1	s_2	s_3	RHS	
1	x_1	1	0	$14/5$	$3/5$	0	$1/5$	$31/5$	
0	s_2	0	0	$156/5$	$22/5$	1	$14/5$	$354/5$	
-3	x_2	0	1	$32/5$	$4/5$	0	$3/5$	$58/5$	
	Z_j	1	-3	$-82/5$	$-9/5$	0	$-8/5$	$-143/5$	
	$C_j - Z_j$	0	0	$97/5$	$9/5$	0	$8/5$		

As $C_j - Z_j \geq 0$, optimal solⁿ is

$$x_1 = 31/5, \quad x_2 = 58/5, \quad x_3 = 0$$

$$\text{and } Z_{\min} = -143/5$$

Two phase method

Two phase method is modified version of simplex method that will solve LPP having $\geq$ and $=$ constraints. As the solution of LPP is calculated in two phases, it is known as **Two - Phase simplex method**

Steps

Convert given LPP to standard form as follows.

- 1) RHS should be +ve, if not convert
- 2) If the constraint has $=$, then add artificial variable to LHS
- 3) If the constraint has $\geq$, then subtract surplus variable & add artificial variable to LHS.
- 4) If the constraint has $\leq$, then add slack variable.

Phase I :- Minimize $Z^* = a_1 + a_2 + \dots + a_n$,
where a_i 's are the added artificial
variables with respect to the given
constraint.

Phase II :- Solve the original problem,
starting from the basic feasible solution

Ex:- Maximize $Z = -2x_1 - x_2$

Subject to

$$3x_1 + x_2 = 3$$

$$4x_1 + 3x_2 \geq 6$$

$$x_1 + 2x_2 \leq 4$$

$$x_1, x_2 \geq 0$$

Solⁿ:- Standard form of LPP is

$$\text{Maximize } Z = -2x_1 - x_2 + 0s_1 + 0s_2 + a_1 + a_2$$

$$\text{subject to } 3x_1 + x_2 + a_1 = 3$$

$$4x_1 + 3x_2 - s_1 + a_2 = 6$$

$$x_1 + 2x_2 + s_2 = 4$$

$$x_1, x_2, a_1, a_2, s_1, s_2 \geq 0$$

Phase I :- Minimize $Z^* = a_1 + a_2$

$$\text{s.t. } 3x_1 + x_2 + a_1 = 3$$

$$4x_1 + 3x_2 - s_1 + a_2 = 6$$

$$x_1 + 2x_2 + s_2 = 4, \quad x_1, x_2, a_1, a_2, s_1, s_2 \geq 0$$

	C_j	0	0	0	0	1	1		
	B.V	x_1	x_2	s_1	s_2	a_1	a_2	RHS	Ratio
1	a_1	(3)	1	0	0	1	0	3	$1 \rightarrow \min$
1	a_2	4	3	-1	0	0	1	6	$3/2$
0	s_2	1	2	0	1	0	0	4	4
	Z_j	7	4	-1	0	1	1	9	
	$C_j - Z_j$	-7	-4	1	0	0	0		

↓
max negative

$$R_1 \rightarrow R_1/3, \quad R_2 \rightarrow R_2 - 4R_1, \quad R_3 \rightarrow R_3 - R_1$$

	C_j	0	0	0	0	1	1		
	B.V	x_1	x_2	s_1	s_2	a_1	a_2	RHS	Ratio
0	x_1	1	$1/3$	0	0	$1/3$	0	1	3
1	a_2	0	(5/3)	-1	0	$-4/3$	1	2	$1.2 \rightarrow \min$
0	s_2	0	$5/3$	0	1	$-1/3$	0	3	1.8
	Z_j	0	$5/3$	-1	0	$-4/3$	1	2	
	$C_j - Z_j$	0	$-5/3$	1	0	$7/3$	0		

↓
max negative

$$R_1 \rightarrow R_1 - \frac{1}{5} R_2, \quad R_2 \rightarrow \frac{3}{5} R_2$$

$$1 - 2/5$$

$$R_3 \rightarrow R_3 - R_2$$

$$1/3 + 1/5 \times 4/3$$

C_j	0	0	0	0	1	1		
B.V	x_1	x_2	s_1	s_2	a_1	a_2	RHS	Ratio
0	x_1	1	0	$\frac{1}{5}$	0	$\frac{9}{15}$	$-\frac{1}{5}$	$\frac{3}{5}$
0	x_2	0	1	$-\frac{3}{5}$	0	$-\frac{4}{5}$	$\frac{3}{5}$	$\frac{6}{5}$
0	s_2	0	0	1	1	1	-1	1
Z_j	0	0	0	0	0	0	0	0
$C_j - Z_j$	0	0	0	0	1	1		

Here all $C_j - Z_j \geq 0$ and values of artificial variables are zero. Therefore we can eliminate the column of artificial variables.

Phase II:- original maximization problem

C_j	-2	-1	0	0		
C_{B_i}	B.V	x_1	x_2	s_1	s_2	RHS
-2	x_1	1	0	$\frac{1}{5}$	0	$\frac{3}{5}$
-1	x_2	0	1	$-\frac{3}{5}$	0	$\frac{6}{5}$
0	s_2	0	0	1	1	1
Z_j	-2	-1	$\frac{1}{5}$	0		$-\frac{12}{5}$
$C_j - Z_j$	0	0	$-\frac{1}{5}$	0		

All $C_j - Z_j \leq 0$, optimal solution is obtained.
 $x_1 = \frac{3}{5}$, $x_2 = \frac{6}{5}$ $Z_{\max} = -\frac{12}{5}$

